MIL-C-51098E(EA)
12 February 1981
SUPERSEDING
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1 November 1976

MILITARY SPECIFICATION

CYLINDER, NITROGEN-FILLED, PUNCTURABLE SEAL

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a nitrogen-filled puncturable seal cylinder.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-N-411	-	Nitrogen, Technical.
QQ-S-781	-	Strapping, Steel, Flat and Seals.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-C-843	-	Cushioning, Material, Cellulosic.

MILITARY

MIL-B-117	-	Bag and Sleeve, Interior Packaging.

FSC 4230

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

C5-51-376 - Cylinder, Nitrogen-Filled, Puncturable Seal.
TDPL 5-51-376 - Cylinder, Nitrogen-Filled, Puncturable Seal.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material and components.

3.1.1 Material. All materials cited on Drawing TDPL 5-51-376, shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawing.

3.1.2 Design. The cylinder shall be in accordance with Drawing C5-51-376.

3.2 Gas charge. The cylinder shall be filled with 4.2 to 4.8 grams (g) of nitrogen and tested for gas charge as specified in 4.4.4.1.

3.3 Performance.

3.3.1 Burst strength. The filled cylinder (see 3.2) shall not burst or rupture when heated to 100° ± 5° C for at least 15 minutes when tested as specified in 4.4.4.2.

3.3.2 Rupture. The filled cylinder (see 3.2) shall not fail at the cap weld (bulge, crack, or tear loose) and the body shall rupture (split open) but not splinter or fragment when heated as specified in 4.4.4.3.

3.3.3 Hydrostatic burst strength. The filled cylinder supplied shall have a content pressure equal to one-half or less than one-half the cylinder hydrostatic pressure before rupture, this assures a 2:1 safety factor. This content pressure is computed when confining 4.8 grams of nitrogen at 160°F (4.4.4.5).

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3.4 Workmanship. The cylinder shall be free from contamination (foreign matter, oil, grease or internal or external rust) and damage such as dents, deformation, burrs or rough or sharp edges. Prior to painting, the cylinder shall be free from contamination, pit marks or cracks and shall show no evidence of bulging.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lot of cylinder, nitrogen-filled, puncturable seal. A preproduction lot of 100 each nitrogen-filled cylinders and an additional 100 empty cylinders shall be manufactured using the same materials, equipment, and processes that will be used during regular production.

4.3.2 Sampling. Sampling of filled cylinders shall be conducted at random in accordance with table I in the sequence listed. A sample of 22 empty cylinders shall be selected at random for test in accordance with 4.3.3.2(c).

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive tests.

(a) Cylinder nitrogen-filled. Each filled cylinder shall be examined and tested for all drawing requirements and workmanship. The acceptance number is zero.

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TABLE I. Sampling

Filled Cylinder Assembly Requirement	Quantity		Inspection Procedure
	Lot	Sample	
Burst strength	100	100	4.4.4.2
Gas charge	100	10	4.4.4.1
Rupture	100	10	4.4.4.3

(b) Cylinder, nitrogen-filled. Each filled cylinder shall be examined and tested in accordance with the classification of defects (4.4.3.3), the AQL's do not apply. The acceptance number is zero.

(c) Internal volume. Using a volumetric displacement method, determine the internal volume of the empty cylinders designated (4.3.3.2(c)) for hydrostatic burst strength test (4.4.4.4). Cylinders shall conform to exterior envelope dimensions of drawing C5-51-376 for this volume determination.

4.3.3.2 For destructive tests.

(a) Gas charge. The ten sample cylinders shall be tested in accordance with 4.4.4.1. If one cylinder fails the test, the lot shall be rejected.

(b) Rupture. The ten sample cylinders shall be tested in accordance with 4.4.4.3. If one cylinder fails the test, the lot shall be rejected.

(c) Hydrostatic burst strength. The 22 sample empty cylinders (see 4.3.2) shall be tested in accordance with 4.4.4.4. If one cylinder fails the test, the lot shall be rejected.

4.3.4 Acceptance/rejection criteria. The preproduction lot shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. A lot shall consist of cylinders from no more than one shift's production.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

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4.4.2.2 For destructive tests. Sampling for each destructive test shall be conducted in accordance with MIL-STD-105, inspection level S-3.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. The sample cylinders and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests. The filled sample cylinders shall be tested in accordance with 4.4.4.1, 4.4.4.3 and 4.4.4.6 using AQL of 1.00%. The empty sample cylinders shall be tested in accordance with 4.4.4.4. The acceptance number is zero.

4.4.3.3 Classification of defects.

(a) Cylinder, nitrogen-filled, puncturable, seal (Drawing C5-51-376).

Categories	Defects	Acceptance standards
Major:	100 percent inspection	
101	Burst strength failure	4.4.4.2
Major:	AQL 1.0 percent defective	
102	Eccentricity excessive	
103	Diameter (0.735) incorrect	
104	Diameter (0.348) incorrect	
105	Length of (0.348) diameter undersize	
106	Overall length incorrect	
107	Depression excessive	
108	Coined cap protrudes beyond (0.348) diameter	
109	Marking missing, illegible or incorrect	
110	Workmanship (see 3.4)	
Minor:	AQL 4.0 percent defective	
201	Protective finish missing or incorrect	

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4.4.3.3 Classification of defects (continued).

(b) Packaging inspection (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 4.0 percent defective	
	101	Quantities in unit, intermediate or shipping containers incorrect
	102	Containers incorrect
	103	Contents loose in containers
	104	Container closure incorrect
	105	Strapping loose or missing
	106	Container torn, broken, crushed or wracked
	107	Box reinforcement incorrect
	108	Marking missing, illegible or incorrect
Minor:	AQL 6.5 percent defective	
	201	Cushioning material missing or incorrect

4.4.4 Tests.

4.4.4.1 Gas charge. After completing the burst strength test (4.4.4.2), the filled cylinder shall be air cooled, cleaned, and dried, and then weighed to the nearest 0.01 gram (g). The cylinders shall then be punctured and the contents captured for analyses (see 4.4.4.6.1). After the nitrogen is exhausted, reweigh to the nearest 0.01 g to determine the weight of the nitrogen. The punctured cylinders shall then be used for analysis (see 4.4.4.6.2).

CAUTION: At 100° C the internal pressure will be approximately 5000 lbs per square inch gage (psig).

4.4.4.2 Burst strength. Submerge the filled cylinder in boiling water for at least 15 minutes. Record any failures and the point of failure, e.g. at the cap weld or the side of the cylinder (see 6.3).

4.4.4.3 Rupture. Place the filled cylinder in an oven and increase temperature until cylinder rupture occurs. Examine the cylinder for conformance to 3.3.2 (see 6.3).

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4.4.4.4 Hydrostatic burst strength. Using a suitable fixture, connect the empty cylinder to a source of hydrostatic pressure. Over a period of not less than 20 seconds, gradually increase to the designated pressure (twice the pressure determined in 4.4.4.5). Maintain that pressure for at least 5 seconds, then gradually increase the pressure until the cylinder ruptures. Record the rupture pressure.

4.4.4.5 Determination of content pressure. The content pressure when confining 4.8 grams of nitrogen at 160°F shall be computed by applying:

$$\frac{4343}{\text{Internal Volume (cubic inches)}} = \text{psi}$$

4.4.4.6 Cylinder contamination.

4.4.4.6.1 Gas analysis. The gas captured in 4.4.4.1 shall be analyzed for compliance with C5-51-376 and Federal Specification BB-N-411. As an alternate, the sample cylinders selected for gas analysis can be separate from paragraph 4.4.4.1 tests (gas charge); however, all of the cylinders must be selected from the same lot (see 4.3.1 and 4.4.1).

4.4.4.6.2 Cylinder internal inspection. The internal surfaces of the sample cylinders previously used in 4.4.4.1 shall be examined for foreign matter in accordance with C5-51-376. The procedure for this examination shall be approved by the Government prior to the preproduction inspection and shall be included in the contractor's quality assurance procedures. As an alternate, the sample cylinders selected for hydrocarbon content tests can be separate from paragraph 4.4.4.1 tests (gas charge); however, all of the cylinders must be selected from the same lot (see 4.3.1 and 4.4.1).

5. PACKAGING

5.1 Unit and intermediate packing (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. Five cylinders shall be packaged in a box conforming to PPP-B-566 or PPP-B-636, type, style and class optional, having inside dimensions of 3-11/16 inches by 3/4 inch by 3-1/4 inches. The box shall be closed in accordance with its respective specification and sealed within a barrier bag conforming to style A, class C, type optional of MIL-B-117, having inside dimensions of 6 by 6 inches length and width.

5.1.1.2 Intermediate packing. Twenty unit packages (100 cylinders), packaged in accordance with 5.1.1.1, shall be packaged together in a fiberboard box conforming to W5C, style RSC, of PPP-B-636, having inside dimensions of 9-1/4 inches by 9-1/4 inches by 3-3/4 inches, length, width and depth respectively. The box opening shall be on the 9-1/4 inches by 9-1/4 inches face. The box shall be closed in accordance with the appendix to PPP-B-636.

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5.1.2 Level C. The cylinders shall be packaged in accordance with 5.1.1, except the intermediate package container may be domestic class and the barrier bag may be omitted. When necessary to restrict free movement of the unit pack in the intermediate container, cushioning conforming to PPP-B-843 cellulosic material shall be used.

5.2 Packing. Levels shall be as specified (see 6.2).

5.2.1 Level A. The cylinders, packaged in accordance with 5.1.1, shall be packed five intermediate packs (500 cylinders) in a wood box conforming to style 4, class 2, 0-50 load limit for type 1 or 2 load for overseas shipment, grade A of PPP-B-621. The inside dimensions of the box shall be 21-3/4 inches by 9-3/4 inches by 9-3/4 inches, length, width and depth respectively. As an alternate the box may conform to class 3, style 2 or 3 of PPP-B-585. Boxes shall be closed and strapped in accordance with the respective box specification. Strapping shall conform to class B of QQ-S-781.

5.2.2 Level B. The cylinders shall be packed in accordance with 5.2.1, except that boxes may be grade B of PPP-B-621 and class 2 of PPP-B-585. Strapping may be class A of QQ-S-781.

5.2.3 Level C. The cylinders, packaged in accordance with 5.1.2, shall be packed, five packages (500 cylinders), in a fiberboard box conforming to grade V3c, style RSC, of PPP-B-636, having inside dimensions of 21-3/4 inches by 9-3/4 inches by 9-3/4 inches length, width and depth respectively. The opening for loading the box shall be on the 21-3/4 inches by 9-3/4 inches face. The box shall be closed in accordance with the appendix to PPP-B-636 and reinforced with non-metalic banding in accordance with the appendix.

5.3 Marking. In conjunction with information furnished in the contract or order, marking shall be as specified herein.

5.3.1 Interior packaging. Unit and intermediate packs shall be marked with the appropriate identification consisting of the National Stock Number, nomenclature, quantity and unit of issue. Also the level of packaging, date packaged, purchase order or contract number in accordance with MIL-STD-129.

5.3.2 Packing. The shipping containers shall be marked in accordance with Drawing D5-51-377.

6. NOTES

6.1 Intended use. The cylinder covered by this specification is intended for use in pressurizing the M11 decontaminating apparatus.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) The level of packaging and packing required.

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6.3 Precautions. Tests specified in 4.4.4.3 should be performed in a hot air oven with a safety door latch. For tests specified in 4.4.4.3, the cylinder should be heated until cylinder rupture occurs. Precautions should be taken in tests 4.4.4.2 and 4.4.4.4 to prevent ruptured cylinders from injuring personnel. WARNING - The above three tests are dangerous and personnel should be notified and precautions taken accordingly (see 4.4.4.1).

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement. (See QSTAG 235, 1 Sep 72 and Amendment-1, 27 Nov 78). When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

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